ADHIPARASAKTHI COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

AI-Driven Quiz System with Blockchain-Based Result Protection on Network

Project Members

RISHI ARAVIND P (510122104033)

SAKTHISWAMY B (510122104039)

Under the Guidance of

NARAYANAN S, M.E..,

ASSOCIATE PROFESSOR/CSE

ABSTRACT

The existing quiz and examination systems used in schools and colleges mainly depend on centralized digital platforms. In these systems, teachers manually prepare questions and conduct exams online or in computer labs. Objective questions such as multiple-choice questions are evaluated automatically, but descriptive answers require manual correction. This increases teachers' workload and causes delays in publishing results. Storing student data and results in centralized databases also creates security risks such as data loss, unauthorized access, and result manipulation. In addition, many existing systems require continuous internet connectivity, which limits their use in institutions with poor network facilities.

To overcome these problems, the proposed system introduces an AI-based quiz management platform that works efficiently in both offline and online environments. The system allows teachers to create quizzes manually, upload text or PDF content, or use AI to automatically generate questions. It supports MCQ, descriptive, and coding-based questions. MCQs are evaluated automatically, while descriptive and coding answers are assessed using AI for fair and accurate results. After evaluation, detailed performance reports and feedback are generated. To ensure security and trust, quiz results are stored using private blockchain technology, making them tamper-proof and reliable. Overall, the proposed system provides a secure, intelligent, and efficient assessment solution for educational institutions.

INTRODUCTION

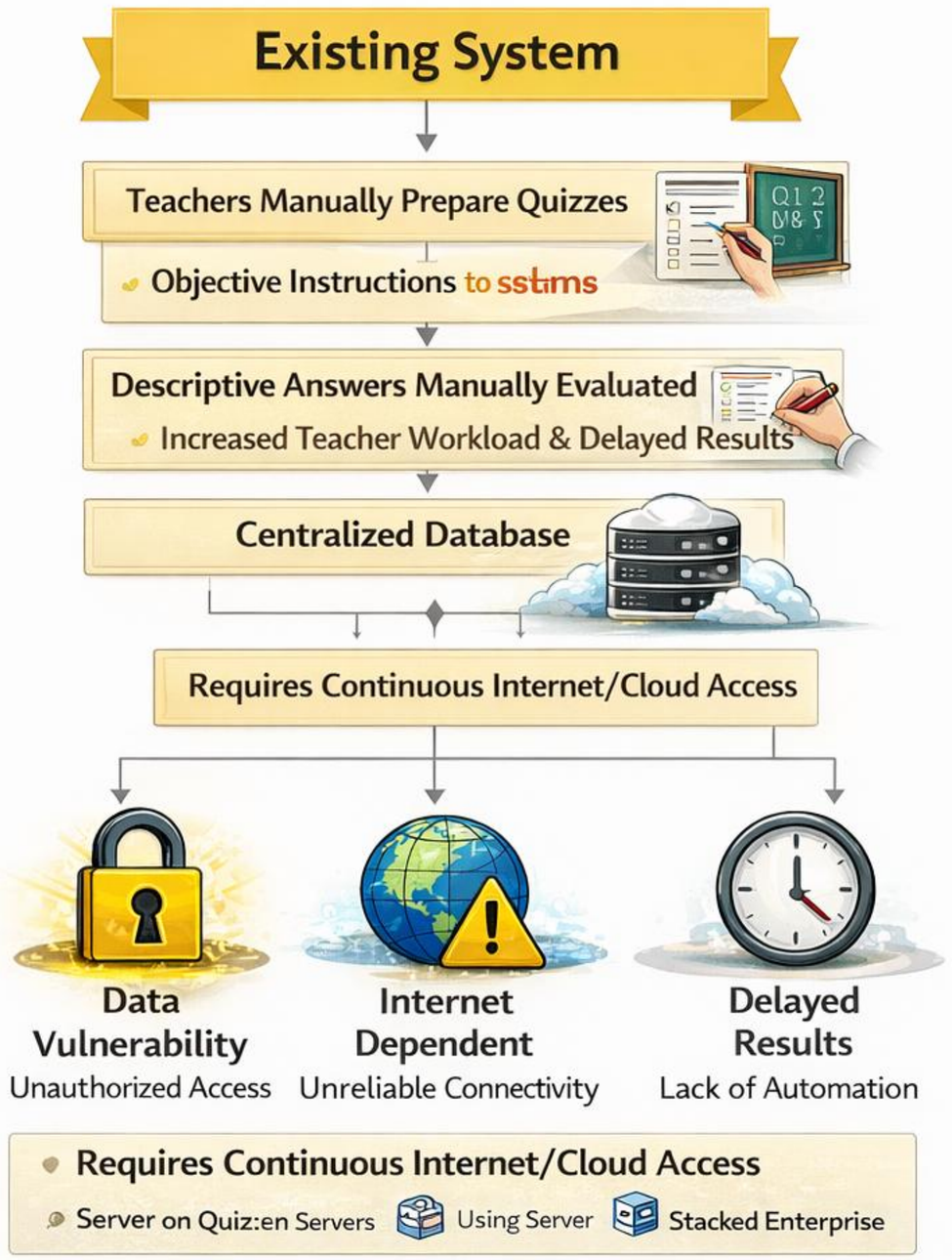
- Quizzes play an important role in evaluating student learning and academic performance.
- Traditional quiz systems involve manual question preparation, evaluation, and result management, making them time-consuming and prone to errors.
- Ensuring the security of quiz results and preventing unauthorized modification remains a major challenge in existing systems.
- The proposed system uses Artificial Intelligence to automate quiz generation and evaluation, reducing manual effort.
- Blockchain technology is employed to securely store quiz results in an immutable and tamper-proof manner.
- The system operates entirely within a local network, ensuring data privacy and making it suitable for institutions with limited internet access.

LITERATURE SURVEY

YEAR	TITLE OF PAPER	AUTHORS	DESCRIPTION
2023	A Proposed Model for Improving the Reliability of Online Exam Results Using Blockchain	Mohamed Abdelsalam et al.	Proposes a blockchain-based model (integrated with Moodle) where questions, answers, and exam results are written to a decentralized ledger to guarantee integrity and traceability of assessments.
			PROBLEM: Conventional online exam systems store results in centralized databases that can be hacked or altered, undermining the reliability and trustworthiness of exam outcomes.
2023	Online quiz implementation using blockchain technology for result tampering prevention using token	Elena-Carmen Tentea, Valeriu Manuel Ionescu	Describes an online quiz platform that stores quiz results on a blockchain and uses cryptographic tokens to access them, ensuring tamper-proof storage and verifiable academic records.
			PROBLEM: Existing online quiz platforms are vulnerable to result tampering and unauthorized modification because they rely on centralized database storage without immutable logging.
2024	A Proposed Model for Improving the Reliability of Online Exam Systems (hybrid / secure architecture)	A. Al Dhanhani et al.	Presents an architecture for online examinations that emphasizes secure communication, reliability, and integration with institutional infrastructure, suitable for deployment over controlled networks.
			PROBLEM: Many online exam systems depend heavily on public internet and basic security measures, leading to outages, latency, and security/privacy risks for institutions.

EXISTING SYSTEM

The Existing System is shown as block diagram



LIMITATIONS OF EXISITNG METHOD

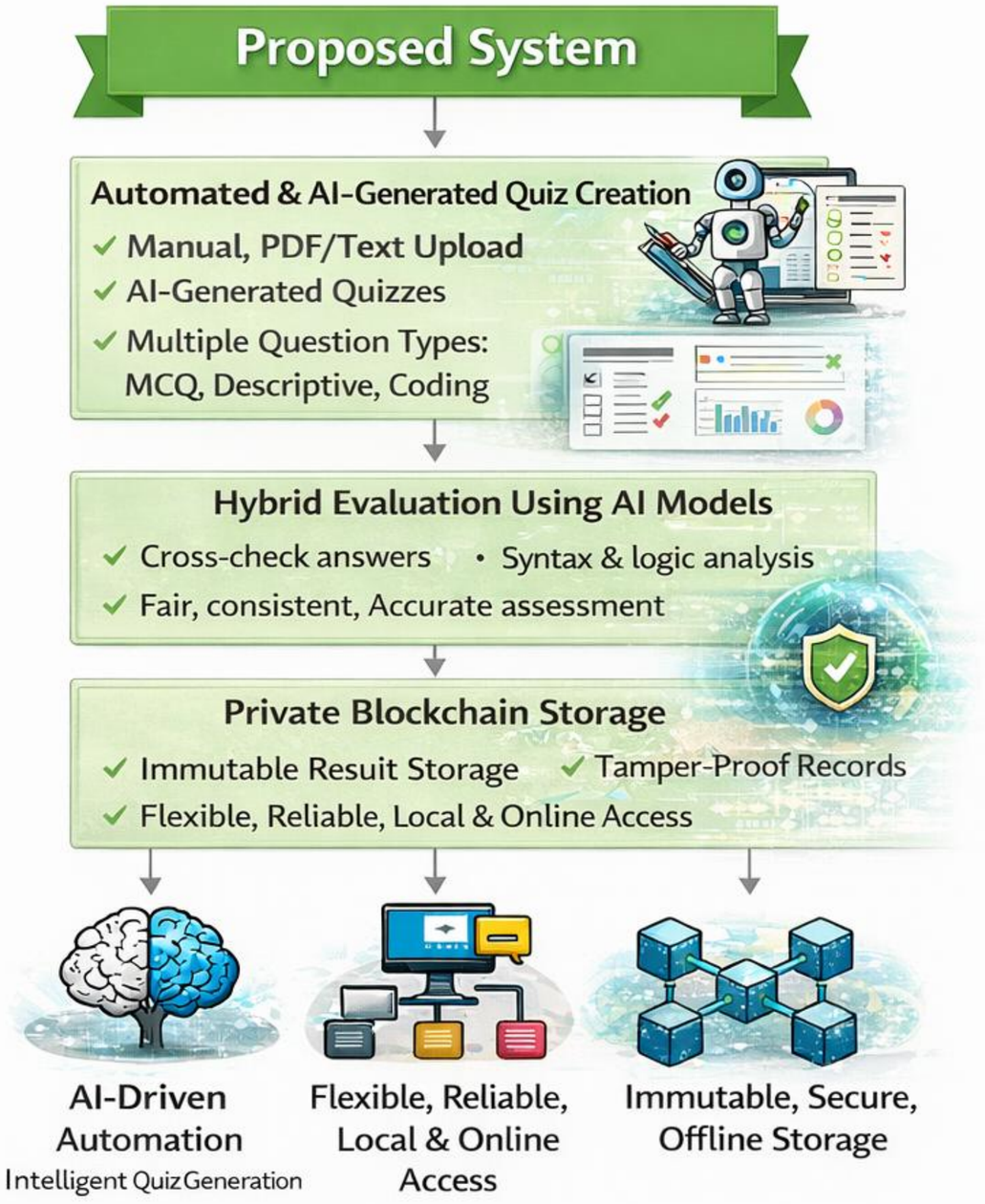
1. Time-Consuming Process
2. Delayed results
3. Error-prone
4. Question leakage
5. No instant feedback
6. Low Security and Fairness
7. Manual Attendance Tracking
8. Lack of Randomization
9. No Personalized Feedback
10. No Auto-Generated Reports
11. High Operational Cost
12. Weak Analytics and Reporting
13. Limited interactivity
14. Low Student Engagement
15. Manual record-keeping
16. Answer copying / cheating

PROBLEM STATEMENT

- Existing quiz systems are time-consuming due to manual question preparation and answer evaluation.
- Manual evaluation increases teacher workload and the possibility of errors.
- Quiz results in conventional systems are vulnerable to unauthorized modification, reducing reliability.
- There is a need for a system that automates quiz creation and evaluation to save time and effort.
- Secure, tamper-proof storage of quiz results is required to ensure data integrity and trustworthiness.
- The proposed system uses Artificial Intelligence for automation and blockchain technology for secure result storage.

PROPOSED SYSTEM

The Proposed System is shown as block diagram



SOFTWARE REQUIREMENTS SPECIFICATION

COMPONENTS	REQUIREMENTS
Frontend	Flet(Flutter)
Backend	Python + FastAPI
Database	SQLite
Authentication	JWT (JSON Web Token)
Network	• Local Area Network (LAN) - Local • Cloudflare Tunnel - Public/Global Access
AI Module	Local LLM (Ollama)
Tools Used	VS Code, Postman, Local Host

REQUIREMENTS SPECIFICATION JUSTIFICATION

1. Flet (Flutter) - [Frontend Tool]

Flet is a Python framework built on Flutter that allows developers to create interactive, real-time web, desktop, and mobile applications using only Python. It simplifies UI development by eliminating the need for separate frontend code, enabling faster app development with a single codebase.

2.1. Python - [Backend Tool]

Python is a high-level, interpreted programming language known for its simplicity and readability. It is widely used for backend development, data science, AI/ML, automation, and scripting due to its rich ecosystem of libraries and strong community support.

2.2. FastAPI - [Backend Tool]

FastAPI is a modern, high-performance web framework for building APIs using Python. It is designed for speed and ease of use, supports asynchronous programming, and automatically generates interactive API documentation using OpenAPI and Swagger UI.

3. SQLite - [Database]

SQLite is a lightweight, serverless, self-contained relational database management system. It stores data in a single file and is widely used for small to medium-scale applications due to its simplicity, reliability, and zero-configuration setup.

4. JWT - [Authentication]

JSON Web Token (JWT) is a compact, secure token used to transmit information between parties, commonly for authentication and authorization, ensuring data integrity without storing session info.

5. [Network]

- **Local Area Network (LAN)** - Provides access **within a local network**, limited to devices connected to the same network.
- **Cloudflare Tunnel** - Provides **public/global access**, allowing devices outside the local network to securely reach your local server.

6. Ollama - [AI Module]

Ollama is a, allowing developers to build aplatform for running **local large language models (LLMs)** on your machinend use AI-powered applications without relying on external cloud services. It provides **fast, private, and customizable AI models** that can process natural language tasks directly on local hardware.

7. [Tools Used]

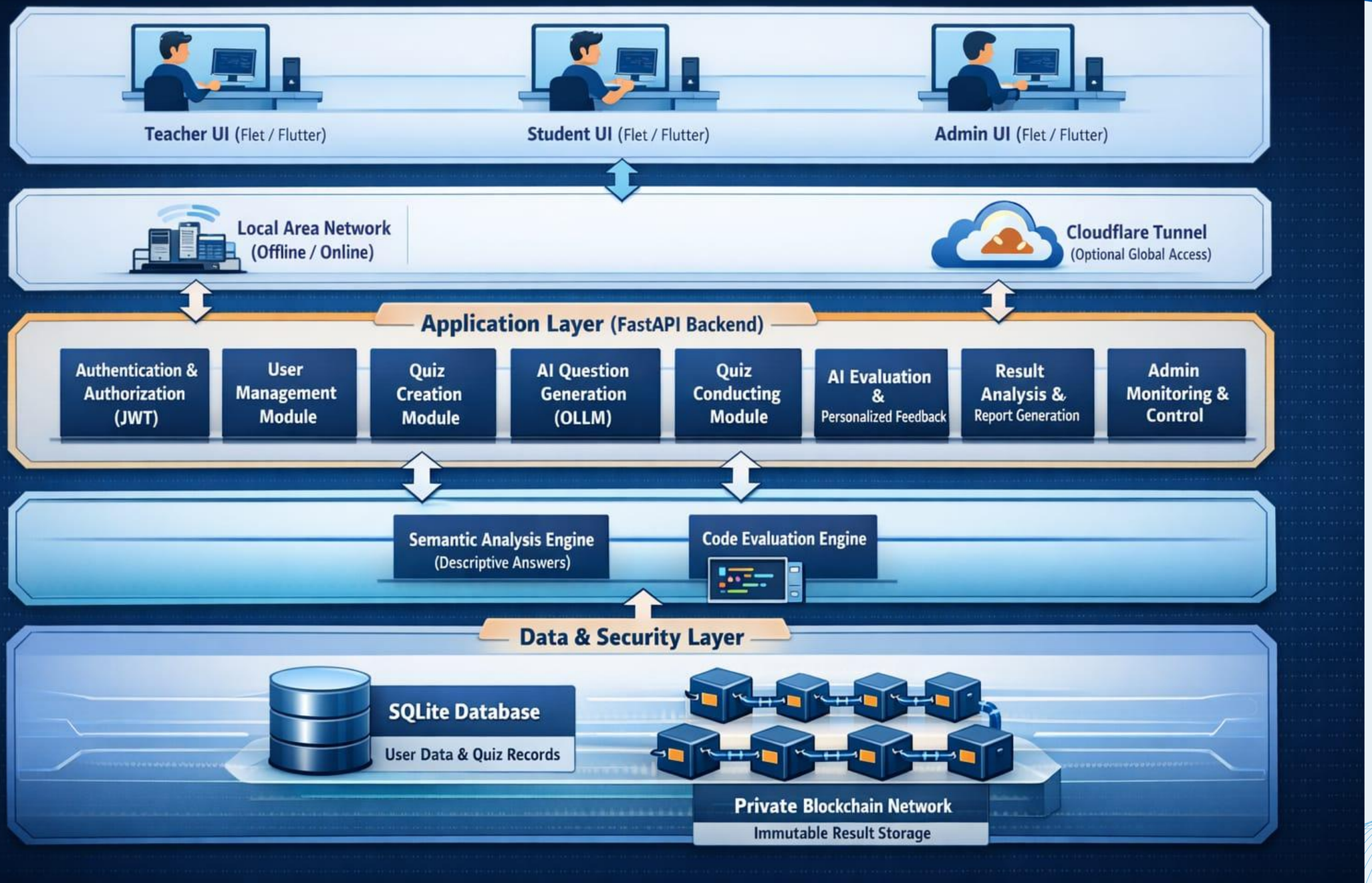
- **Postman** - Used to **test, debug, and interact with APIs**, making it easier to verify endpoints and data responses.
- **Visual Studio (VS)** - The primary **development environment** for writing, editing, and managing code efficiently.
- **Localhost - Hosts the application locally**, allowing testing and development before deploying to a server or cloud.

HARDWARE REQUIREMENTS SPECIFICATION

COMPONENTS	REQUIREMENTS
Processor	Intel i3 or higher
Ram	Minimum 8 GB (16 GB recommended for training AI Models & Backend System Server)
Hard Disk	128 GB SSD or more
Internet	Only required when the quiz want to accessed by globally

ARCHITECTURE

AI-Driven Quiz System with Blockchain-Based Result Protection on Network



DIAGRAM

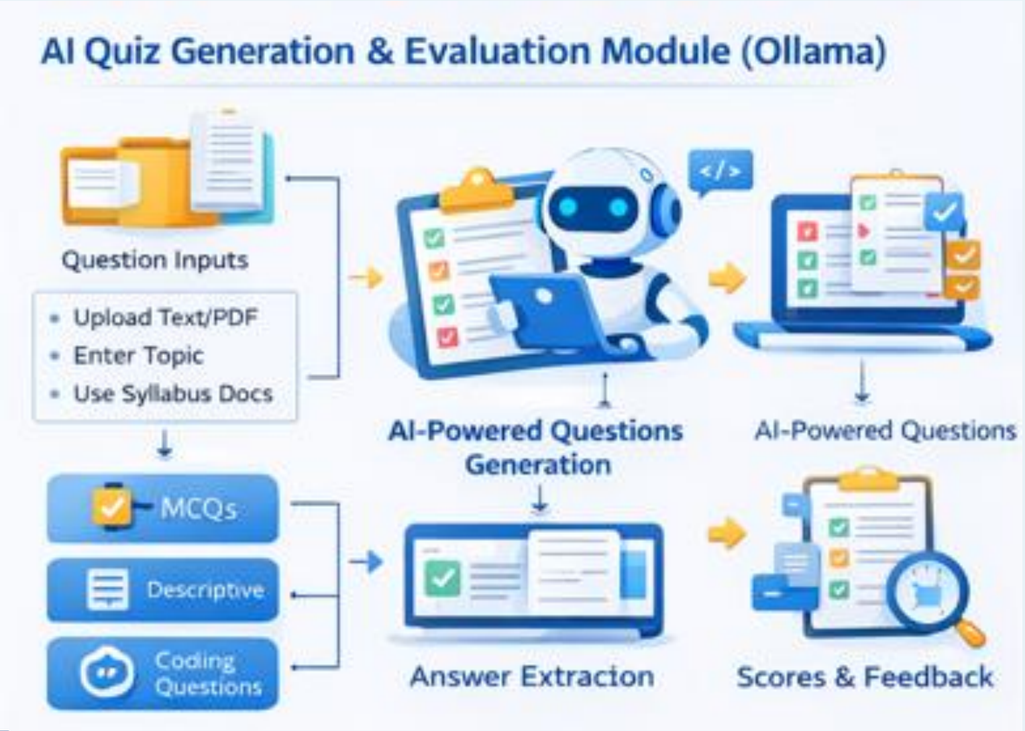
MODUELS DESCRIPTION

User Authentication Module

The User Authentication Module ensures that only authorized users can access the system by securely handling user registration, login, and logout operations. It uses role-based access control to differentiate permissions for students, teachers, and administrators.

JSON Web Tokens (JWT) are used to maintain secure and stateless authentication sessions. This module also safeguards sensitive user information through secure password handling and token validation, ensuring controlled and safe system access.

AI Quiz Generation & Evaluation Module (Ollama)



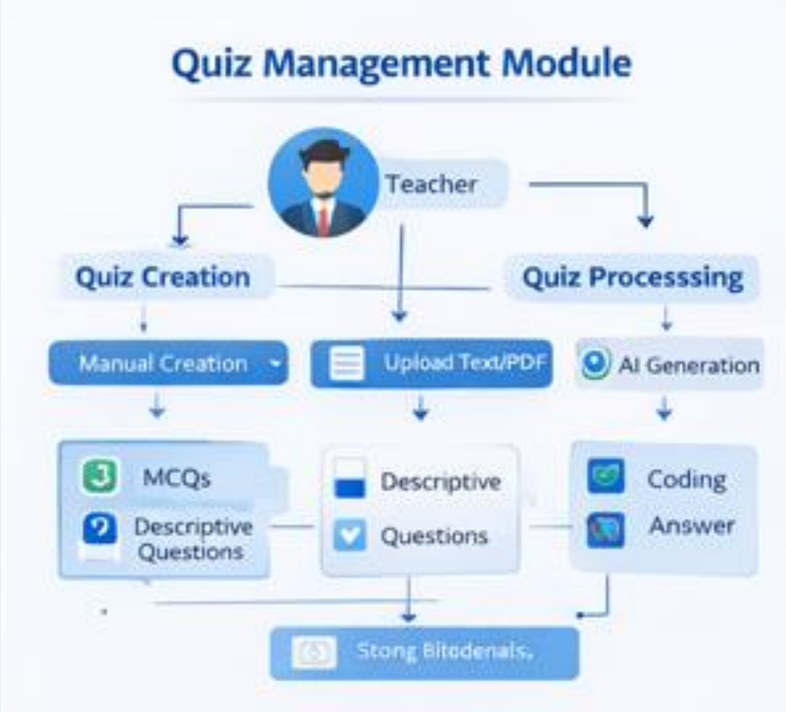
The AI Quiz Generation and Evaluation Module uses local Large Language Models through Ollama to automatically generate quiz questions from topics, syllabus content, or uploaded documents such as PDFs and text files. This reduces manual effort for teachers while ensuring question quality and diversity.

The module also evaluates descriptive and coding answers using AI techniques to provide fair and consistent assessment. Since the AI models run locally, student data privacy is maintained without reliance on external cloud services.

Quiz Management Module

The Quiz Management Module allows teachers to efficiently create, organize, and manage quizzes. It supports different types of questions such as multiple-choice and descriptive questions. Teachers can schedule quizzes, set time limits, and control quiz availability for students.

This module also manages quiz distribution and student submissions. It ensures smooth quiz conduct and collects responses systematically for evaluation, thereby improving the overall assessment process.



Blockchain Result Storage Module

The Blockchain Result Storage Module ensures secure and tamper-proof storage of quiz results using private blockchain technology. Once the evaluation is completed, the results are converted into blocks and stored in an immutable ledger. This prevents unauthorized modification or manipulation of academic records.

By maintaining transparency and data integrity, this module increases trust among students, teachers, and administrators. It also provides a reliable way to verify results in the future.

Network Access Module

The Network Access Module enables the system to function effectively within a Local Area Network (LAN), making it suitable for institutions with limited internet connectivity. It allows students and teachers to access the system within the local network without relying on external servers.

To provide secure external access when required, the module uses Cloudflare Tunnel. This ensures safe remote access without exposing the local server directly to the internet, thereby enhancing system security.

THANK YOU